

Technical Document #570: Cloud Computing - Comprehensive Overview

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Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Kubernetes orchestrates containerized applications across clusters of machines.
- Serverless computing allows developers to run code without managing servers.
- Microservices architecture structures applications as a collection of loosely coupled services.